VLLM 压测合并报告: 2k vs 10k Context

- 生成时间: 2026-03-30 14:58:41
- 接口: `http://127.0.0.1:62272/v1/completions`
- 模型: `Qwen3-32B`
- 输出长度统一为 `1024` token
- 对比轮次:
  - `results_fine`: 输入约 `2049` token, 并发 `5 10 20 40 80 160`
  - `results_ctx10k`: 输入约 `10001` token, 并发 `5 10 20 40 80 160`

核心结论

- `2k` 输入时，最佳点出现在并发 `80`，成功吞吐约 `0.65` req/s，timeout 为 `0%`。
- `10k` 输入时，最佳稳定点下降到并发 `40`，成功吞吐约 `0.34` req/s，timeout 仍为 `0%`。
- 当并发提升到 `80` 时，`2k` 输入仍然稳定，而 `10k` 输入已经 `100% timeout`。
- 这说明在当前部署和 `180s` timeout 口径下，长上下文请求显著压缩了可用并发窗口。

关键对比表

并发	2k Mean(ms)	10k Mean(ms)	2k Timeout%	10k Timeout%	2k req/s	10k req/s
5	53723.67	60502.36	0.00%	0.00%	0.09	0.08
10	58459.94	68501.00	0.00%	0.00%	0.16	0.14
20	67825.09	84188.99	0.00%	0.00%	0.28	0.23
40	83235.00	114460.30	0.00%	0.00%	0.46	0.34
80	117575.98	N/A	0.00%	100.00%	0.65	0.00
160	174480.29	N/A	48.75%	100.00%	0.45	0.00

推荐结论

- 如果业务请求常态接近 `1w` 输入 token，建议生产并发先控制在 `40` 以内。
- 如果业务主要在 `2k` 输入附近，当前配置还能承受到 `80` 左右并发。
- 如需把 `1w` 输入负载推到更高并发，优先检查 KV cache、batch 调度和 timeout 预算，而不是只加压客户端并发。

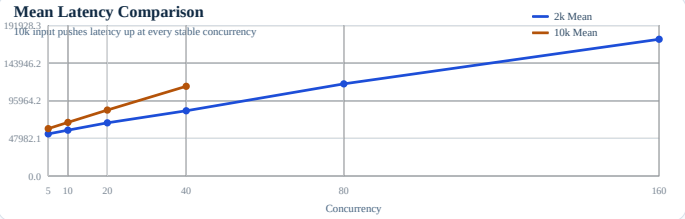
图表

VLLM Benchmark Comparison: 2k vs 10k Input Context

Same endpoint and output length; comparison date 2026-03-30 14:58:41

Mean Latency Comparison

10k input pushes latency up at every stable concurrency

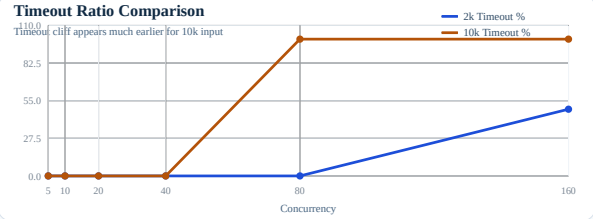


2k vs 10k Summary

并发	2k Mean	10k Mean	2k Timeout%	10k Timeout%	2k re
5	53723.67	60502.36	0.00%	0.00%	0.0
10	58459.94	68501.00	0.00%	0.00%	0.1
20	67825.09	84188.99	0.00%	0.00%	0.2
40	83235.00	114460.30	0.00%	0.00%	0.4
80	117575.98	N/A	0.00%	100.00%	0.6
160	174480.29	N/A	48.75%	100.00%	0.4

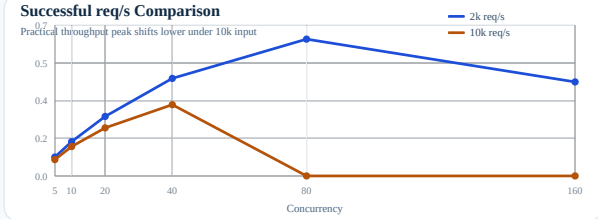
Timeout Ratio Comparison

Timeout cliff appears much earlier for 10k input



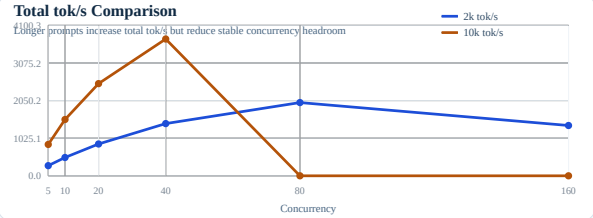
Successful req/s Comparison

Practical throughput peak shifts lower under 10k input



Total tok/s Comparison

Longer prompts increase total tok/s but reduce stable concurrency headroom



Interpretation

1. 2k input remains stable through concurrency 80, while 10k input saturates before 80.
2. At concurrency 40, mean latency rises from 83.24s to 114.46s when input grows from 2k to 10k.
3. At concurrency 80, 2k still has 0% timeout and 0.65 req/s, but 10k becomes 100% timeout.
4. 10k input improves total successful tok/s at low stable concurrency, but loses queueing headroom quickly.
5. For production, 10k input should keep concurrency at or below 40 under this timeout budget.